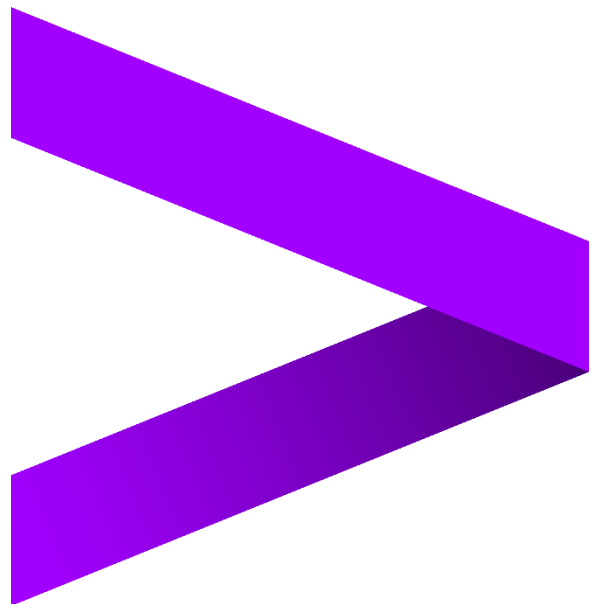


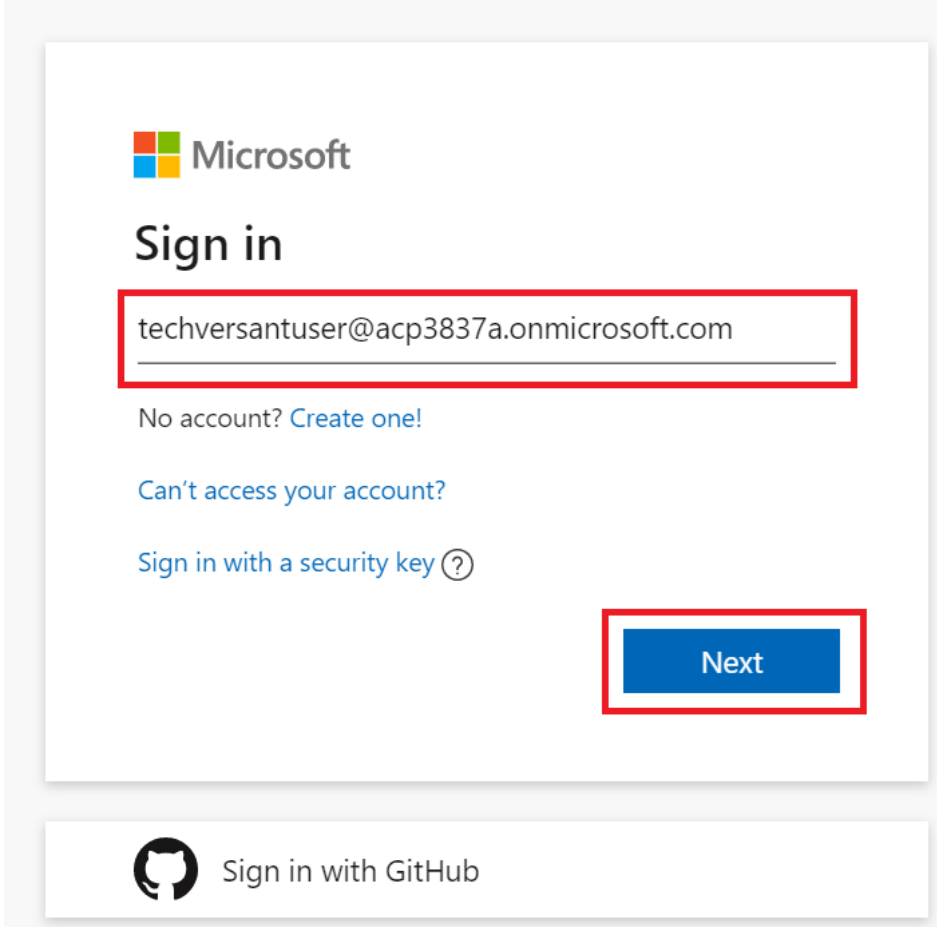
Azure DevOps

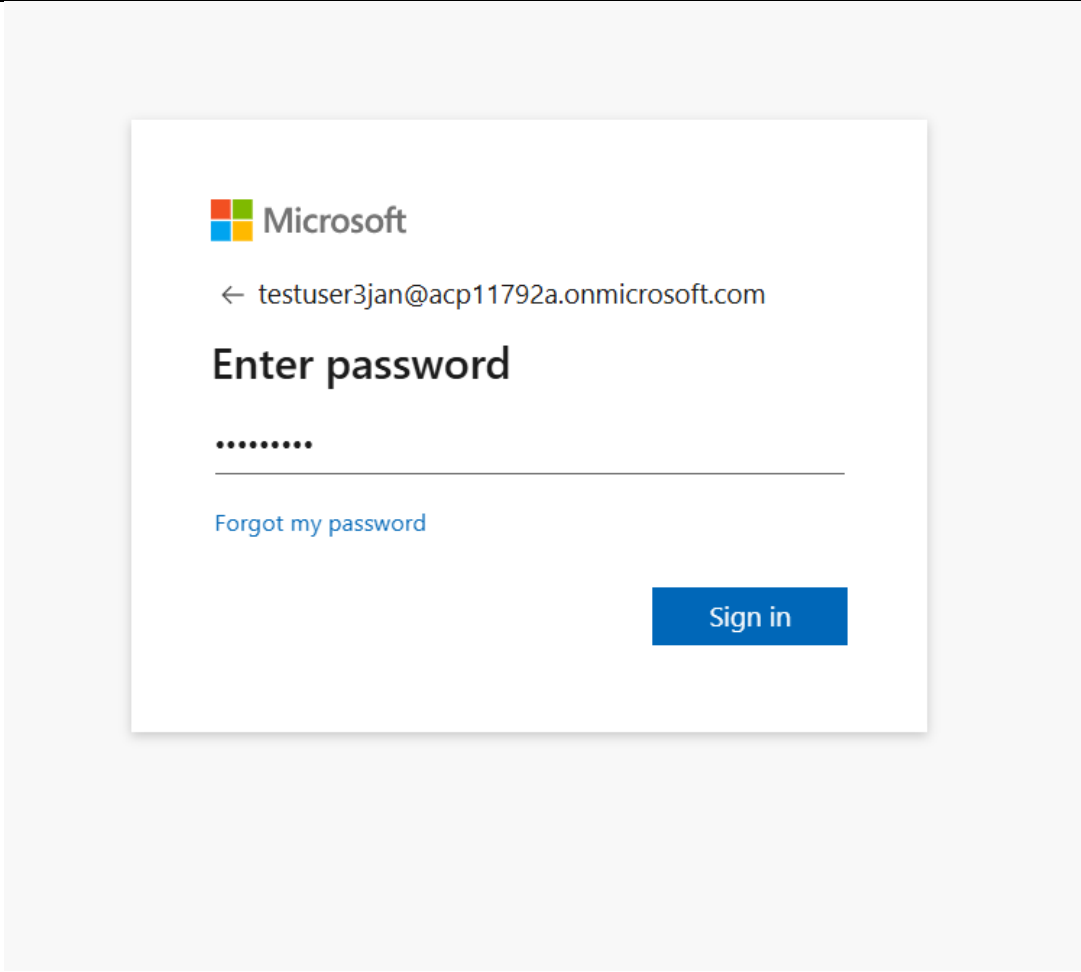
DotNet Case Study Guide

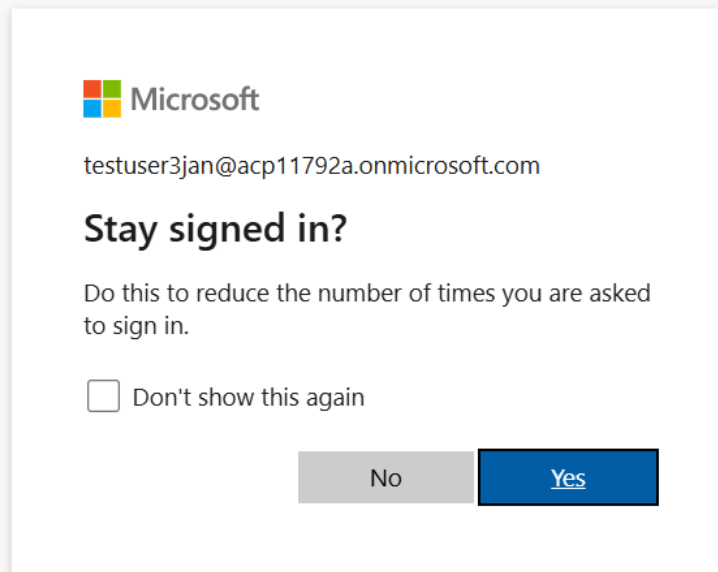


1 AzureDevOps Portal Login

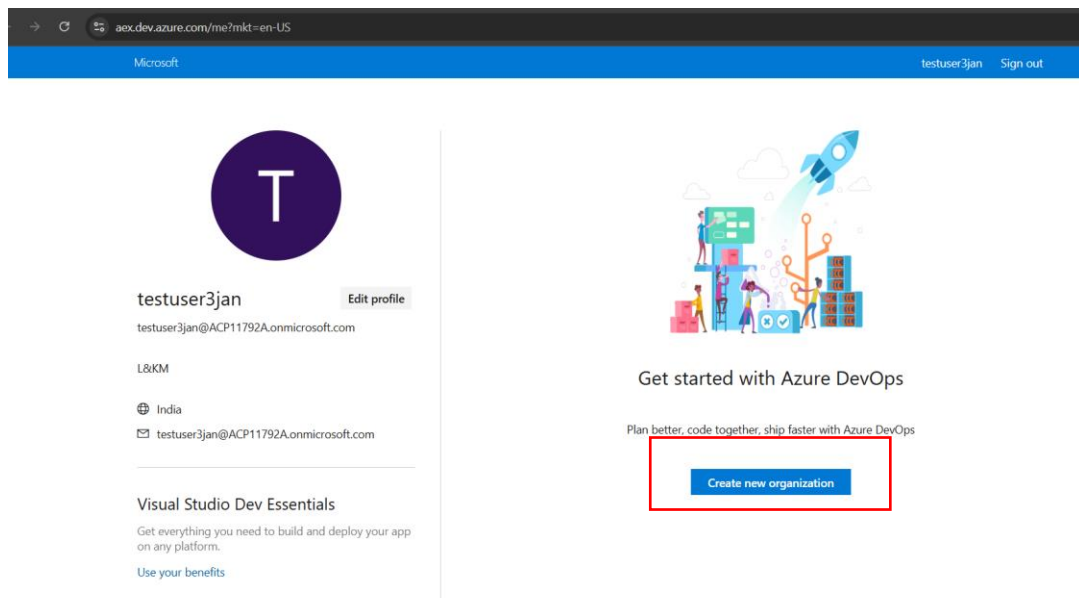
Azure DevOps Portal Login

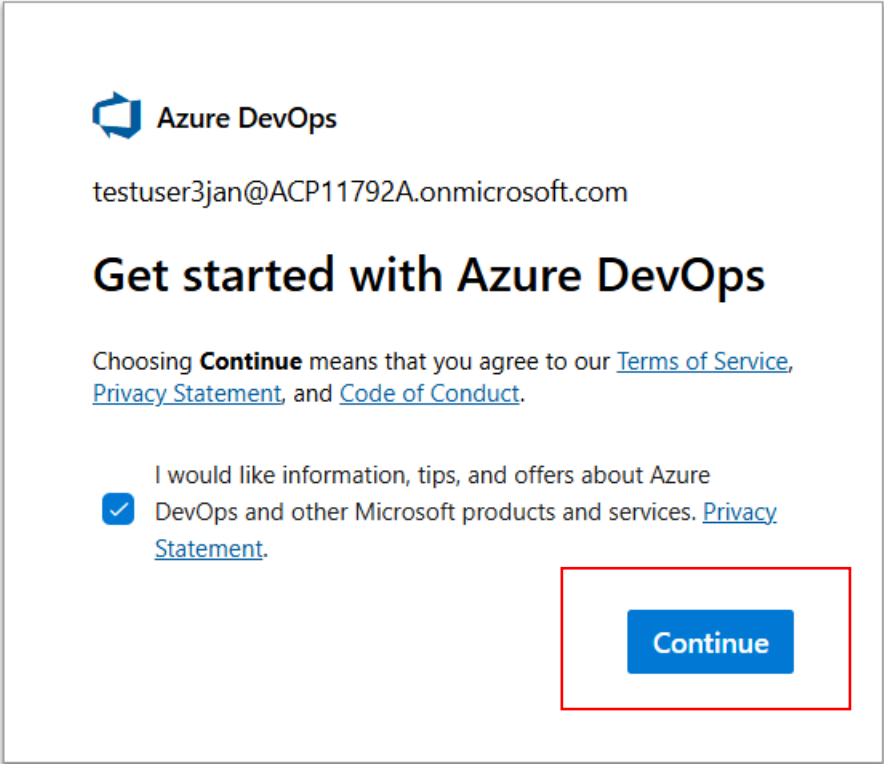
1	Open the chrome browser in incognito mode and navigate to URL https://aex.dev.azure.com/
2	Enter the username provided to you in the login screen. Example username here is “testuserjan3@acp3837a.onmicrosoft.com” and click on Next button. 
3	Enter the password Acc1234\$\$ in the next window.

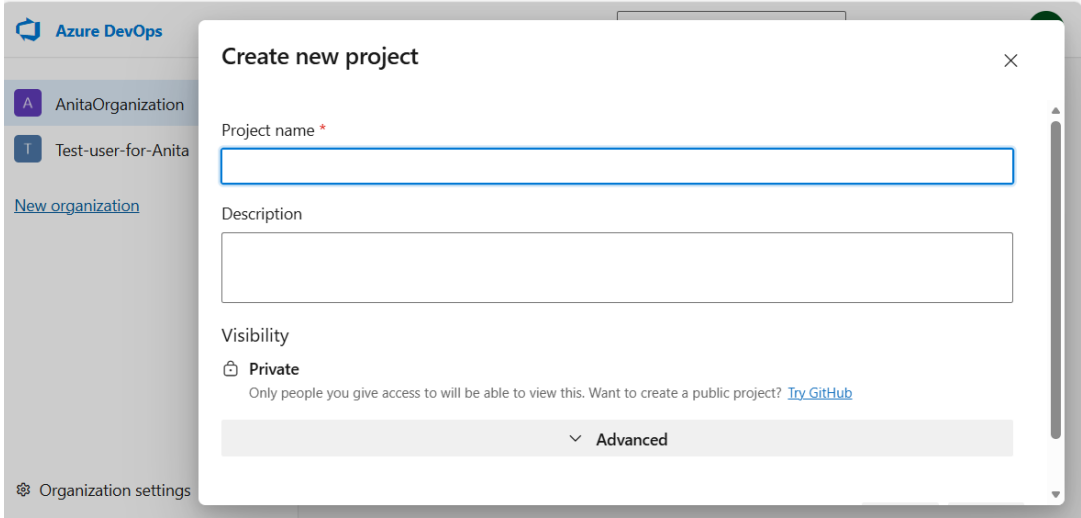
	 A screenshot of the Microsoft login interface. At the top is the Microsoft logo. Below it is a back arrow followed by the email address 'testuser3jan@acp11792a.onmicrosoft.com'. The main heading is 'Enter password'. Below this is a password input field with eight dots. A link 'Forgot my password' is positioned below the input field. A blue 'Sign in' button is located at the bottom right of the login card. The entire card is centered on a light gray background.
4	Click Next and Complete the MFA through authenticator app.
5	In next window you can click yes or no as per your wish.

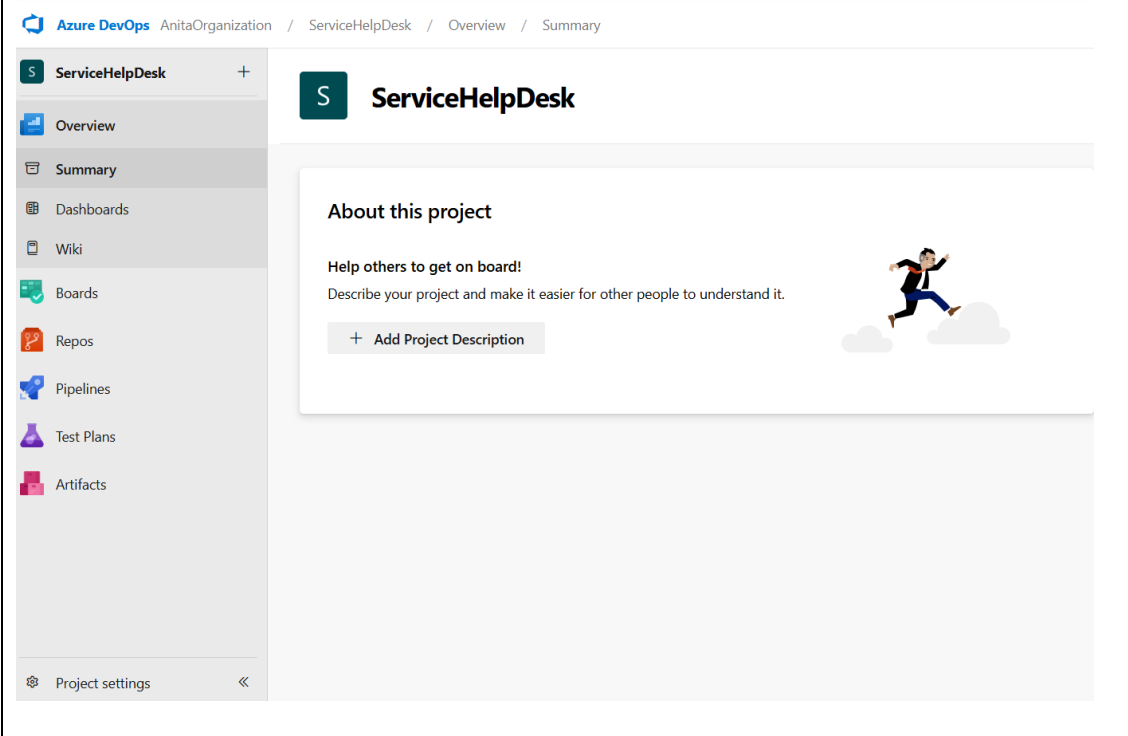
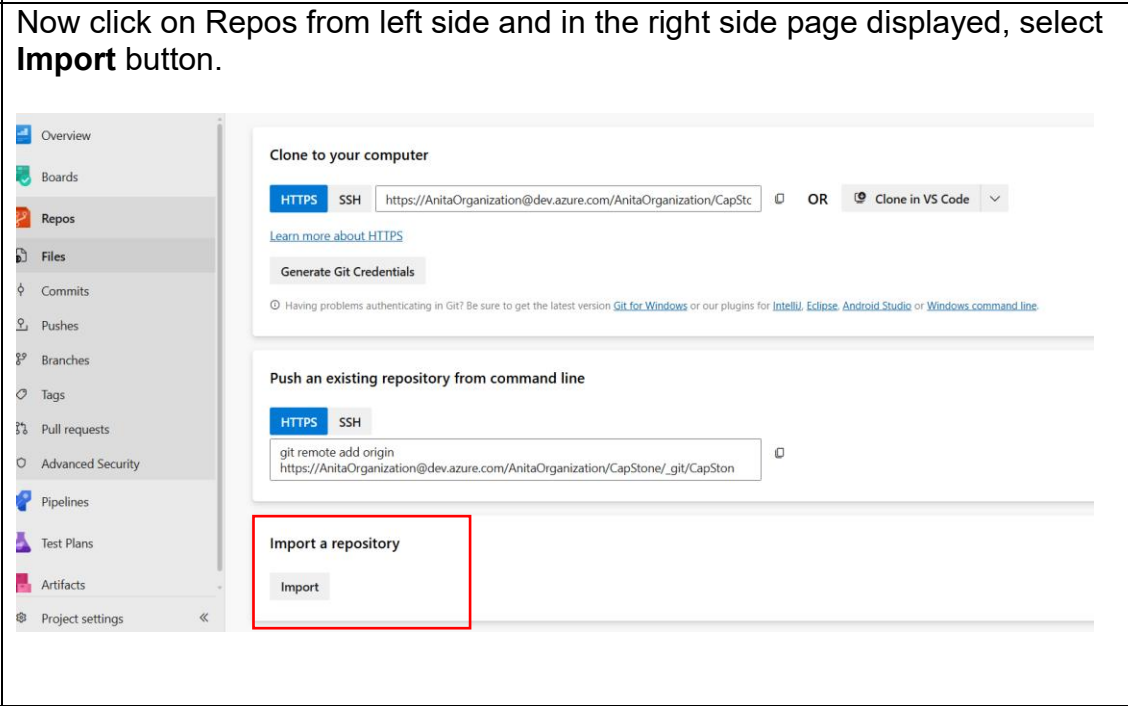


After clicking on Yes you will be navigated to the below page, Click on **Create New organization**



	Now click on Continue 
6	In the page enter the organization name as <yourname>Organization. Example here entered as AnitaOrganization. Then select location as India and enter the captcha if it ask for it.
7	Once click on Continue. Observe the organization is created and highlighted/selected in the left menu. If not selected , select the appropriate organization you have created.

	
	
11	Enter the project name as “ServiceHelpDesk” and select visibility as Private and click on Create project button.
12	Now the project overview page is displayed as shown below

	
13	Now click on Repos from left side and in the right side page displayed, select Import button. 
14	Select Repository type : Git. Enter the clone URL as https://vishnukkallimakula@dev.azure.com/vishnukkallimakula/EnvironmentRepo2/_git/EnvironmentRepo2 And click on Import

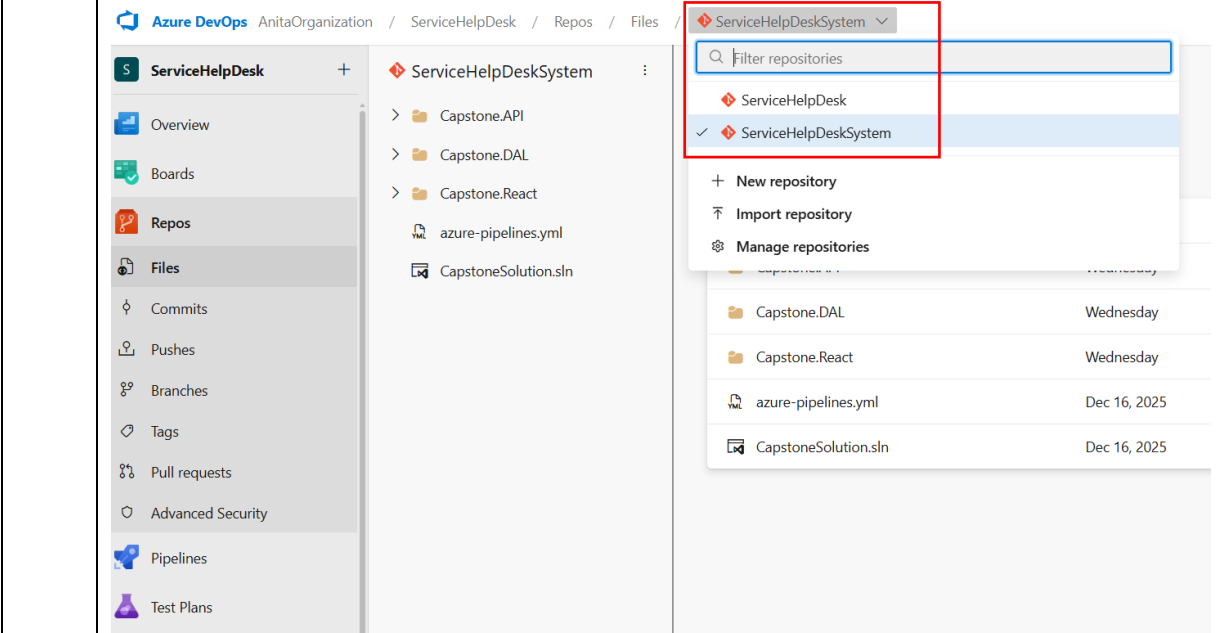
	Import a Git repository × Repository type  Git ▼ Clone URL * https://vishnukkallimakula@dev.azure.com/vishnukkallimakula/Envi ☒ Requires Authentication Username vishnu.k.kallimakula Password / PAT * Cancel Import
15	Wait for few seconds for the import to complete.

16

In the similar way import one more repository.

Now click project name ServiceHelpDesk and select import repository as highlighted below

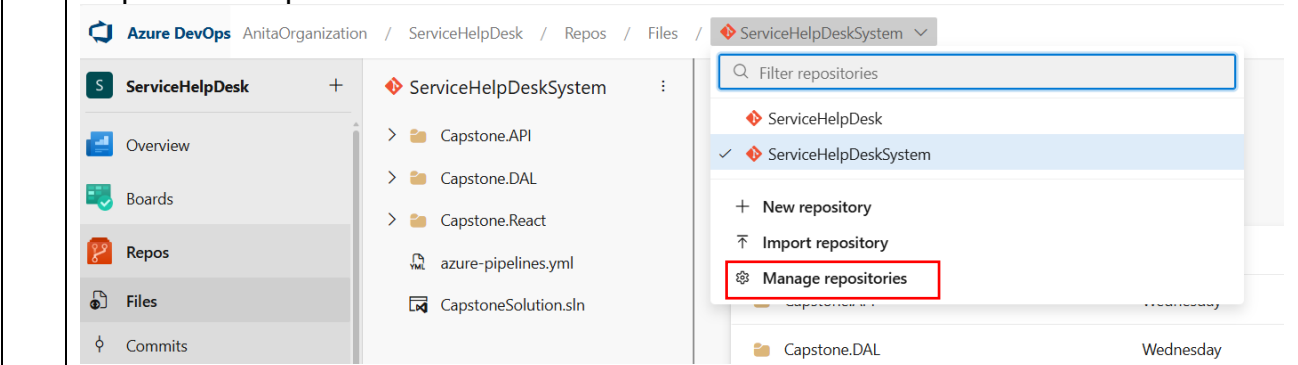
17	Select repo type as Git and enter the clone URL as https://vishnukkallimakula@dev.azure.com/vishnukkallimakula/CapstoneProject1/_git/CapstoneProject1 and click Import Import a Git repository × Repository type  Git ▼ Clone URL * https://vishnukkallimakula@dev.azure.com/vishnukkallimakula/Cap: ☒ Requires Authentication Username vishnu.k.kallimakula Password / PAT * Name * ServiceHelpDeskSystem Cancel Import
18	Observe there are two repositories in your project

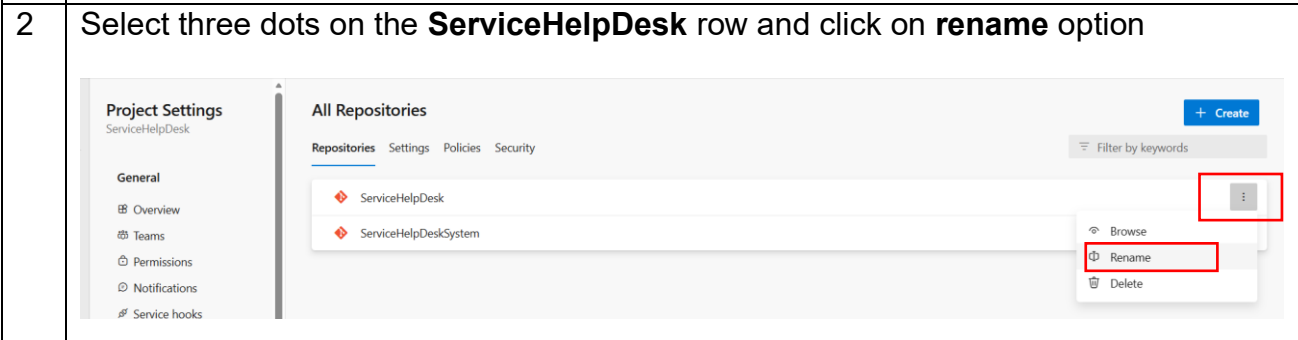


For example here ServiceHelpDeskSystem repo is selected and hence tick mark is present.

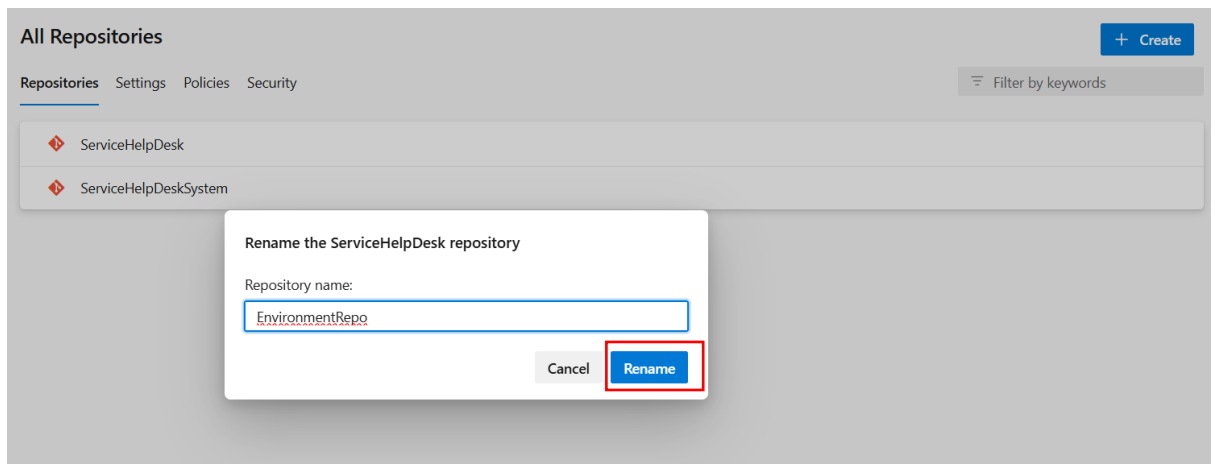
Rename Repo

- Rename one repo to EnvironmentRepo to avoid confusion. For that now click on Repos from left side menu and click on Repositories drop down and select Manage Repositories option.

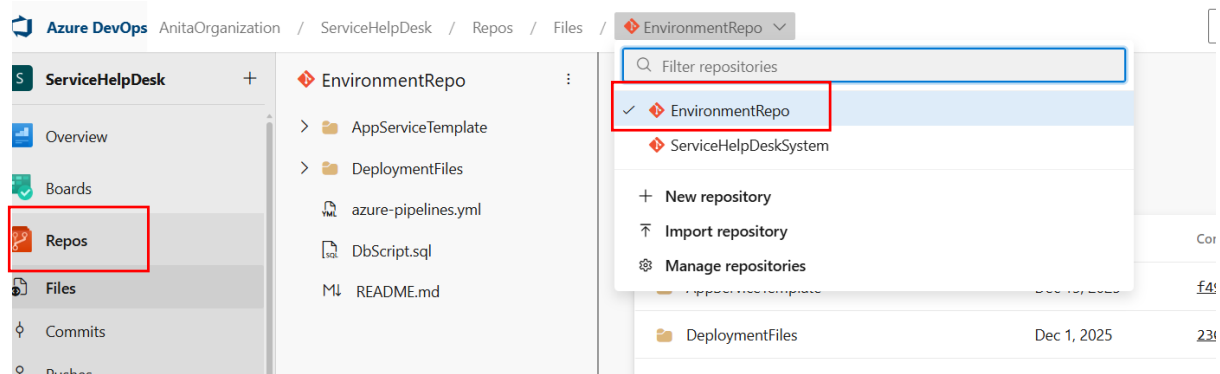

- Select three dots on the **ServiceHelpDesk** row and click on **rename** option



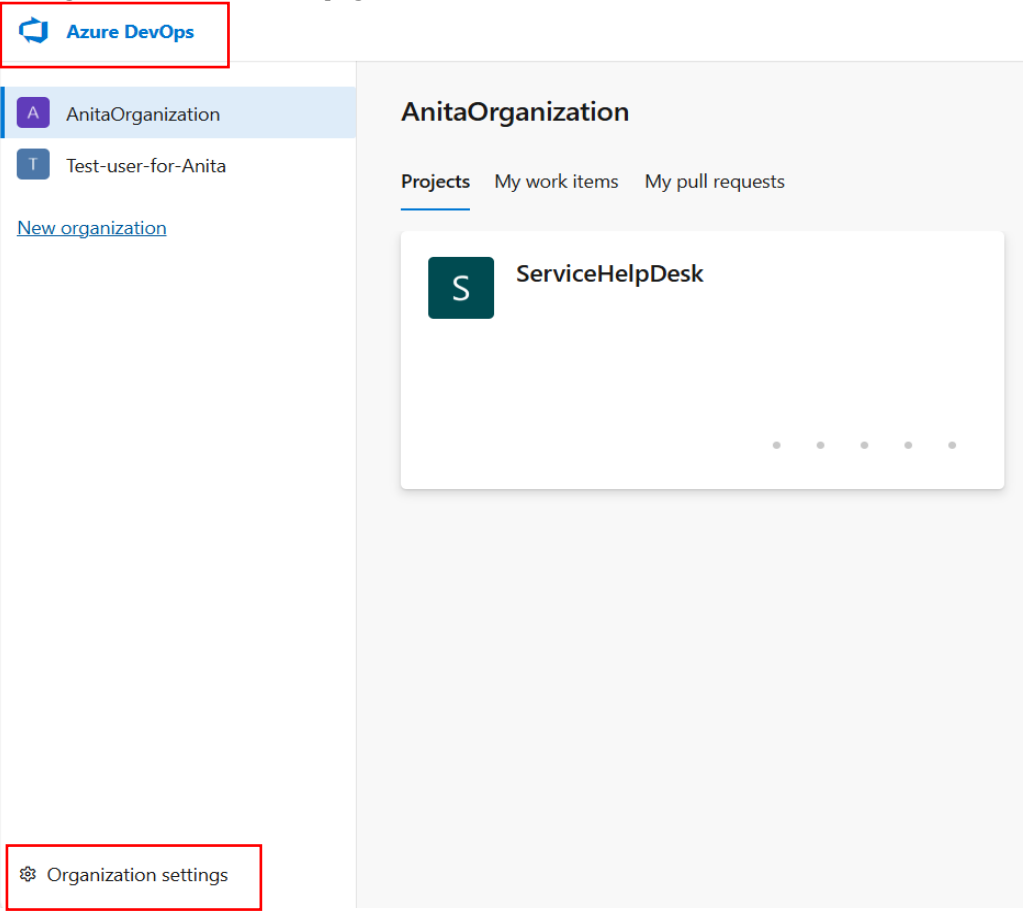
- 3 Delete the old name **ServiceHelpDesk** and give new name as EnvironmentRepo and click on Rename button.



- 4 Now from left hand menu click on Repos and see the EnvironmentRepo name updated and click on the same.



Agent Pool Configuration

1 -	Click on Azure DevOps at the top left, select your organization created and click Organization Settings at the bottom of the page. 
2	Select Agent pool

Add agent pool

Agent pools are shared across an organization.

Pool type.

☐ Managed DevOps Pool

Reduce the effort spent in maintaining custom agents by creating a Microsoft managed pool of scalable agents. [Learn more.](#)

☒ Self-hosted

Create a pool of custom agents hosted on your own infrastructure for maximum control and flexibility. [Learn more.](#)

☐ Azure virtual machine scale set

Create a pool of custom agents based on an Azure Virtual machine scale set hosted in your own Azure subscription. [View configuration instructions.](#)

Name:

AnitaAgentPool

Description (optional):

Markdown supported.

Pipeline permissions:

☒ Auto-provision this agent pool in all projects

Create

Click on **Create**

5

Select the agent pool created.

Agent pools

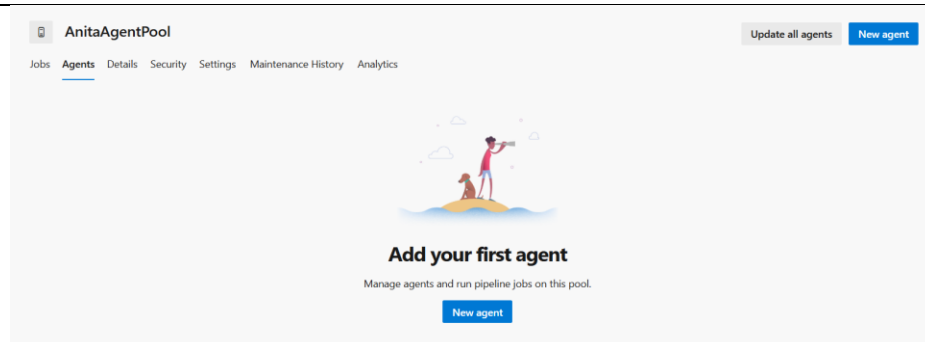
Security

Add

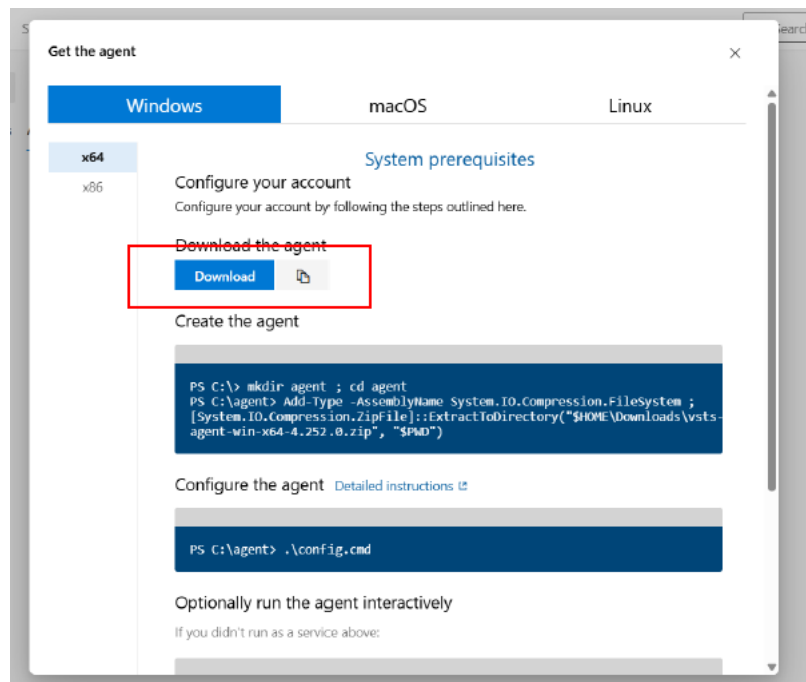
Name	Queued jobs	Running jobs
AnitaAgentPool Test access for Anita	0	1
Azure Pipelines Azure Pipelines		
Default Azure Pipelines		

6

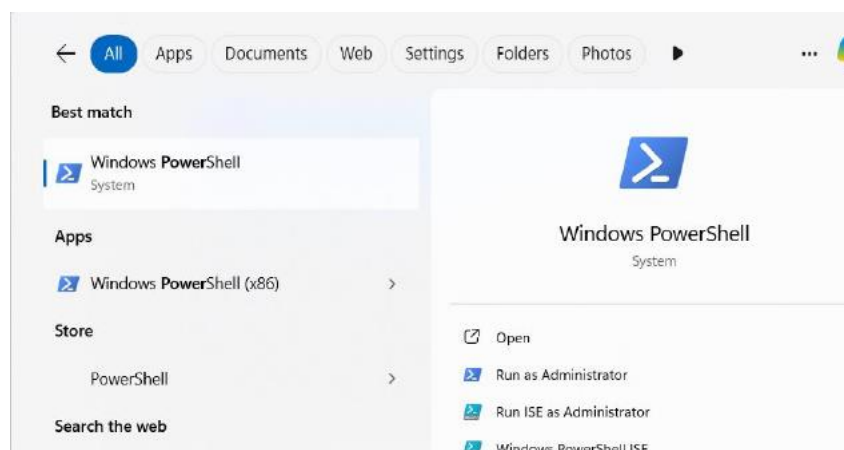
Select **Agents** and click on **New Agent**



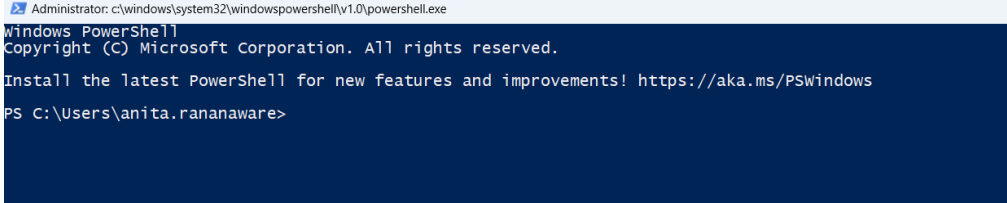
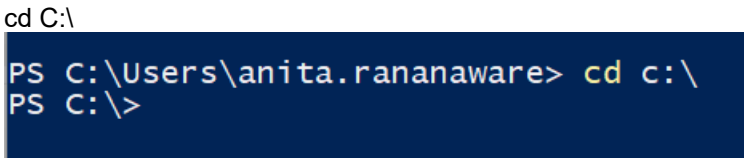
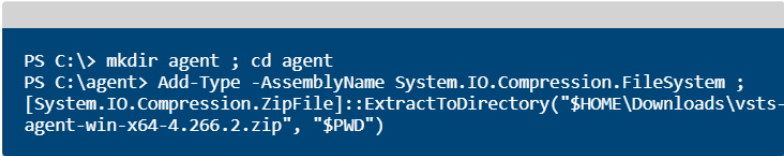
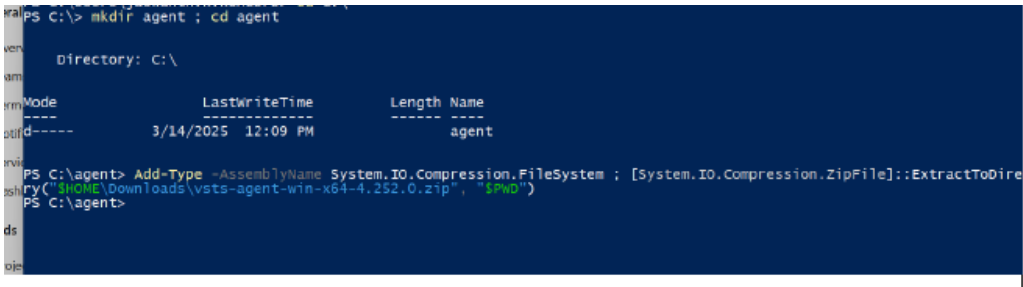
Get the Agent, select **Windows**, Click on Download, agent will be downloaded in to Downloads folder in your machine.



7 After downloading, Open **Powershell** in your machine



Dotnet_Azure DevOps_CaseStudy

	<pre>Administrator: c:\windows\system32\windowspowershell\v1.0\powershell.exe Windows PowerShell Copyright (C) Microsoft Corporation. All rights reserved. Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows PS C:\Users\anita.rananaware></pre> Navigate to C drive; refer commands in below page <pre>cd C:\ PS C:\Users\anita.rananaware> cd c:\ PS C:\></pre> Execute the command from the below page, Create the agent <pre>PS C:\> mkdir agent ; cd agent PS C:\agent> Add-Type -AssemblyName System.IO.Compression.FileSystem ; [System.IO.Compression.ZipFile]::ExtractToDirectory("\$HOME\Downloads\vsts- agent-win-x64-4.266.2.zip", "\$PWD")</pre> Configure the agent Detailed instructions <pre>PS C:\agent> .\config.cmd</pre> Optionally run the agent interactively If you didn't run as a service above: <pre>PS C:\agent> .\run.cmd</pre> That's it! More Information <pre>PS C:\> mkdir agent ; cd agent Directory: C:\ Mode LastWriteTime Length Name ---- - d----- 3/14/2025 12:09 PM agent PS C:\agent> Add-Type -AssemblyName System.IO.Compression.FileSystem ; [System.IO.Compression.ZipFile]::ExtractToDire try("\$HOME\Downloads\vsts-agent-win-x64-4.262.0.zip", "\$PWD") PS C:\agent></pre>
	Run .\config.cmd Provide below details Enter server URL=> Ex URL: https://dev.azure.com/AnitaOrganization/ Ref ss:

Enter authentication type (press enter for PAT) => press **Enter**
Enter personal access token => **<Give pat token here>**
To generate pat token follow below steps:

nt
this pool.

Personal Access Tokens
These can be used instead of a password for applications like Git or can be passed in the authorization header to access REST APIs

Access scope: **Jaswanthor1824** Status: **Active**

Create a new personal access token

Name

testtoken

Organization

AnitaOrganization

Expiration (UTC)

30 days

2/9/2026

Scopes

Authorize the scope of access associated with this token

Scopes

☒ Full access

☐ Custom defined

Create

Cancel

Copy the pat token and save it somewhere for later use.
Enter agent pool (press enter for default) > **<give pool name> Ex: AnitaAgentPool**

Enter work folder (press enter for _work) => press **Enter**
Enter run agent as service? (Y/N) (press enter for N) => Press **Enter**
Enter configure autologon and run agent on startup? (Y/N) (press enter for N) => Press **Enter**

C:\agent>.\config.cmd

agent v4.264.2

(commit 491eca9)

>> Connect:

Enter server URL > https://dev.azure.com/AnitaOrganization/
Enter authentication type (press enter for PAT) >
Enter personal access token > *****
Connecting to server ...

>> Register Agent:

Enter agent pool (press enter for default) > AnitaAgentPool
Enter agent name (press enter for PDC2C-L-4VL18S3) > AnitaAgent
Scanning for tool capabilities.
Connecting to the server.
Successfully added the agent
Testing agent connection.
Enter work folder (press enter for _work) > Anita
2026-01-09 18:51:42Z: Settings Saved.
Enter run agent as service? (Y/N) (press enter for N) >
Enter configure autologon and run agent on startup? (Y/N) (press enter for N) >

C:\agent>

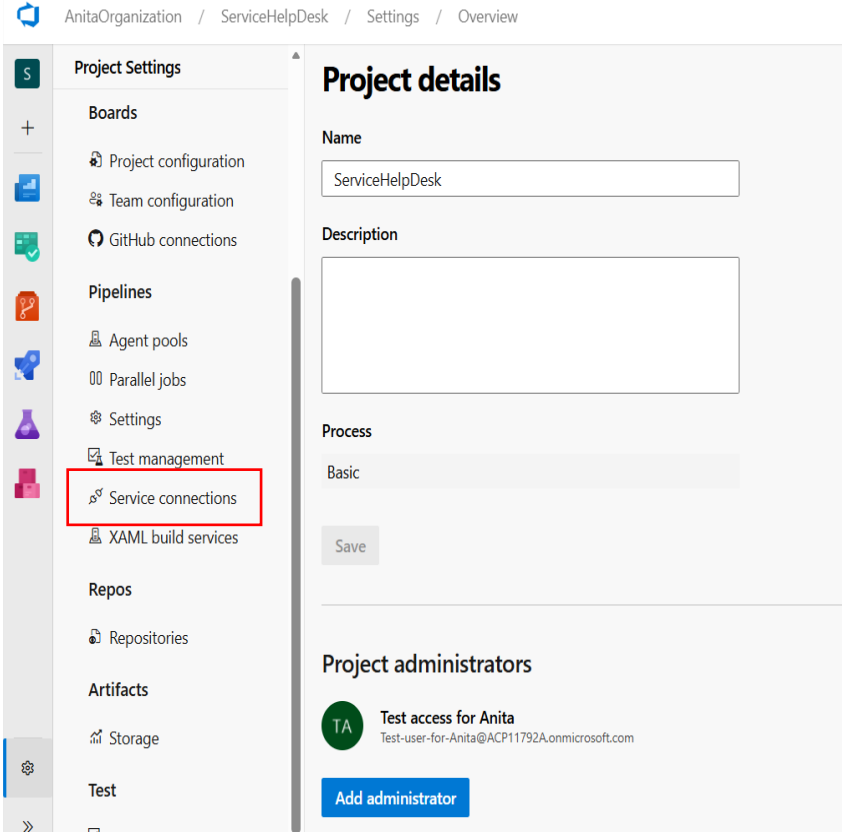
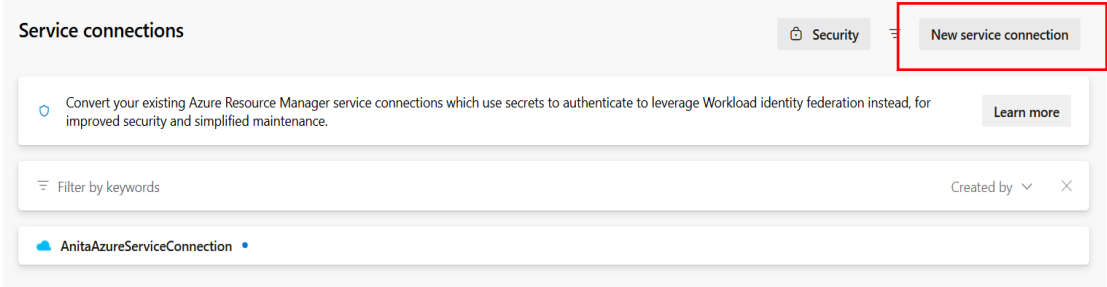
Execute below command .\run.cmd

C:\agent>.\run.cmd

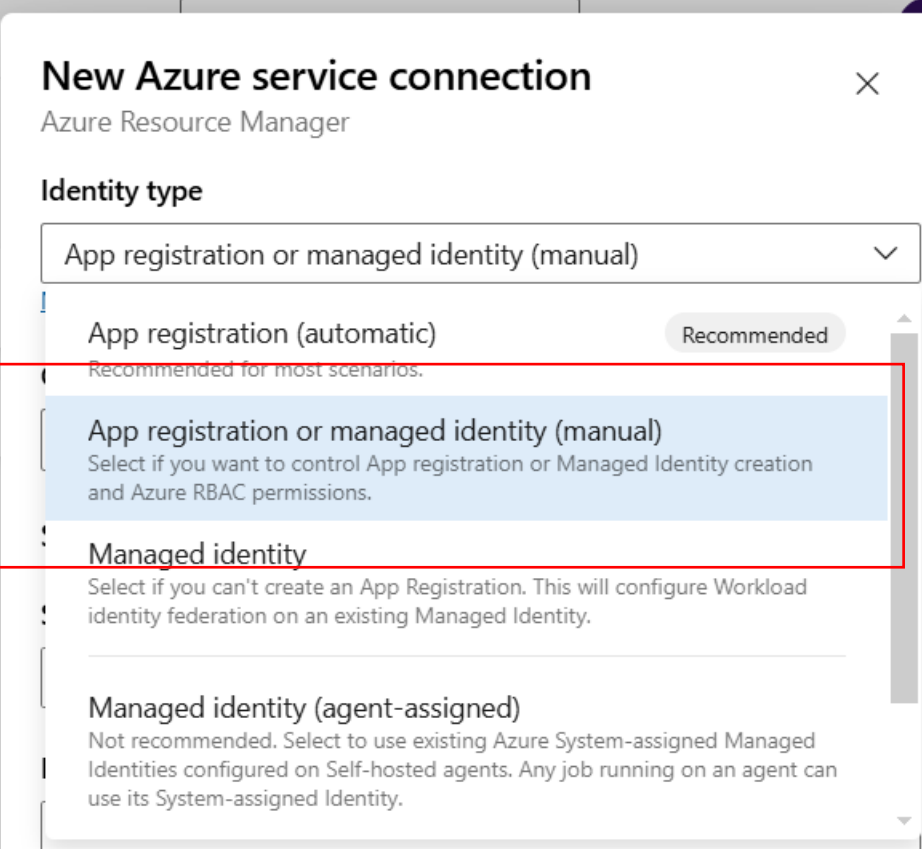
Scanning for tool capabilities.
Connecting to the server.
2026-01-09 18:53:29Z: Listening for Jobs

Service connection set up for Azure Service

1	Go to Project settings, from the left menu, Under Pipelines, click on Service Connections
---	---

	
2	Now in the service connection page, click on New service connection button. 
3	In New service connection page, select Azure Resource Manager option and click on Next

	New service connection Choose a service or connection type Search connection types ☐ Azure Classic ☐ Azure Repos/Team Foundation Server ☒ Azure Resource Manager ☐ Azure Service Bus ☐ Bitbucket Cloud ☐ Cargo ☐ Chef ☐ Docker Host ☐ Docker Registry ☐ Generic Learn more Next
3	Select Identity type App Registration or managed identity (manual) option

	
4	Enter / Select the options as given. Highlighted values are provided to you by cohorts team already. Credential : as Secret (from drop down) Environment : Azure Cloud Scope level : Subscription Subscription Id : <Enter the provided value> Subscription name : <Enter the provided value> Application (client) ID: <Enter the provided value> Directory (tenant) ID: <Enter the provided value> Credential: select Service principal key Client secret: <Enter the provided value> Provide service connection name as <yourname>serviceconnection as shown below After entering Client secret click on Verify and ensure Verification succeeded. Give the service connection name as <yourname>AzureServiceConnection. For example AnitaAzureServiceConnection and same can be given in description. Ensure check box is checked under security and click on verify and save button.

5 Ensure there are two service connections under your project

New Azure service connection

Azure Resource Manager

App registration or managed identity (manual) 

[Need help choosing a connection type?](#)

Credential

Secret 

Warning: using a secret or certificate will require manual rotation and management. We recommend using workload identity federation. [Learn about credential types](#)

Environment

Azure Cloud 

Scope Level

☒ Subscription

☐ Management Group

☐ Machine Learning Workspace

Subscription ID

d1000be4-4abb-4675-b709-04c98058ec17

Subscription name

My Wizard Azure

New Azure service connection

Azure Resource Manager

Application (client) ID

Directory (tenant) ID

Credential

☒ Service principal key ☐ Certificate

Client secret

Verify ✔ Verification Succeeded

Service Connection Name

Description (optional)

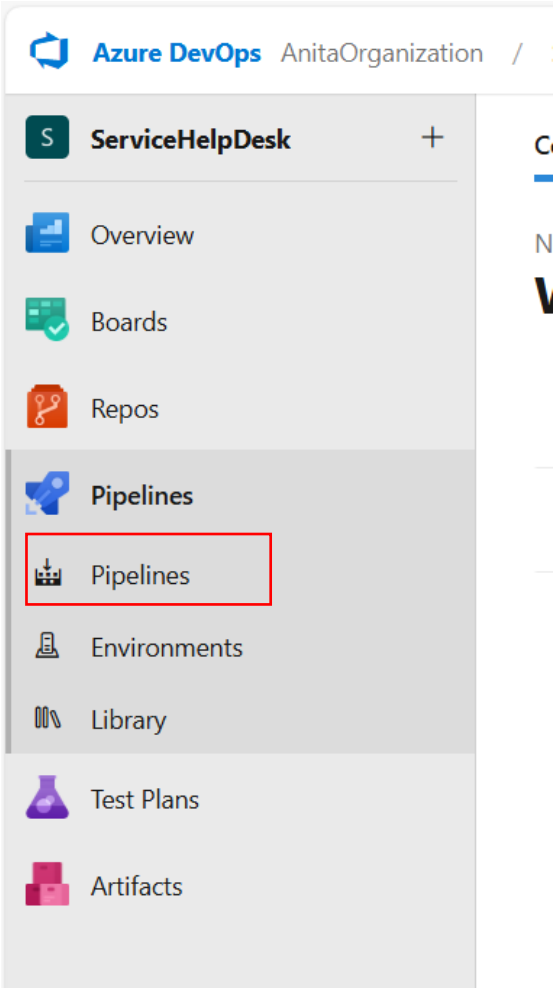
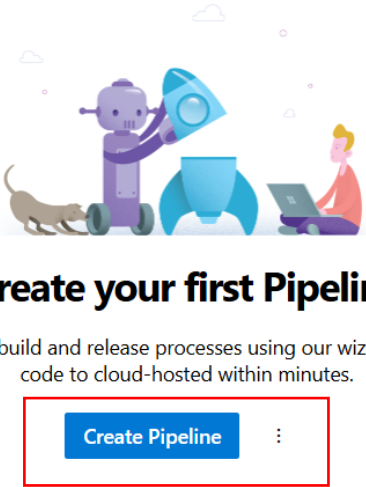
Security

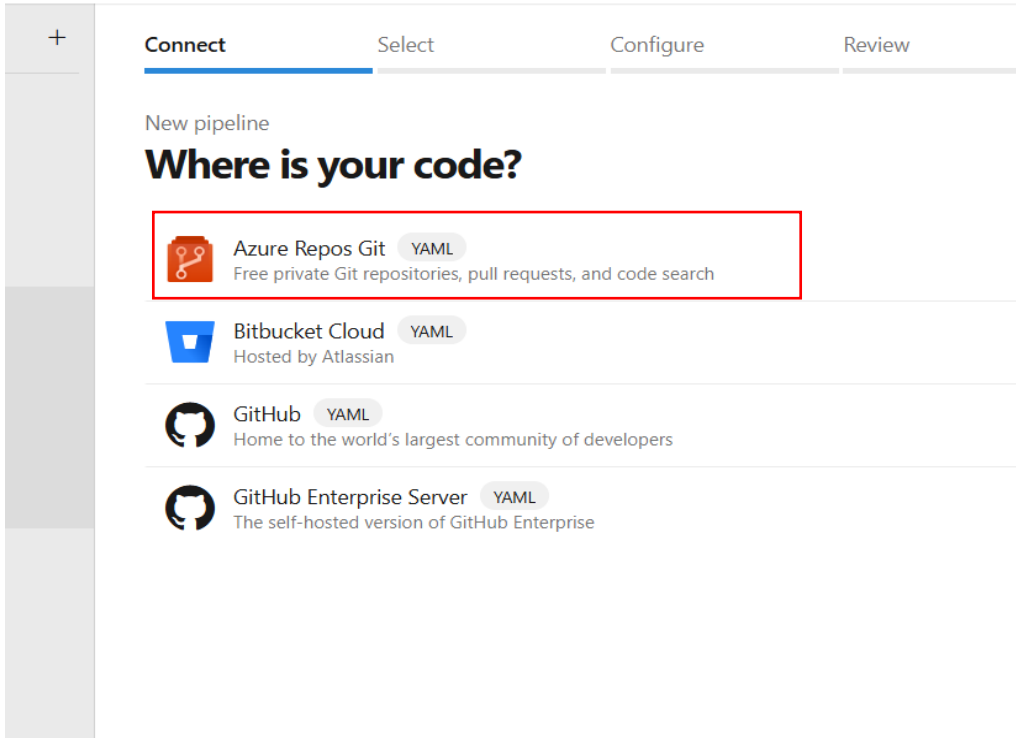
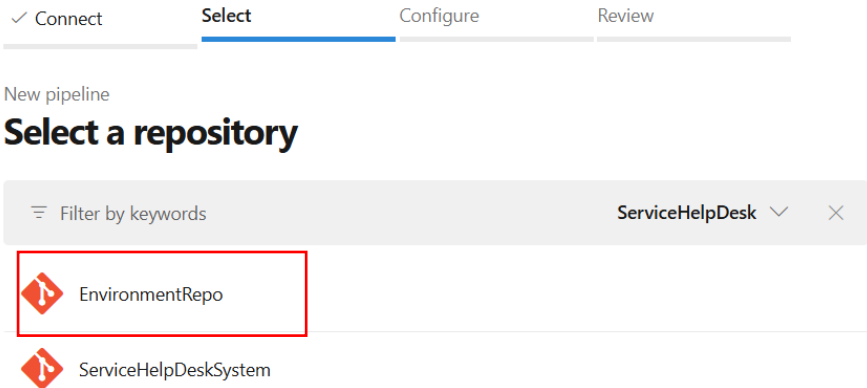
☒ Grant access permission to all pipelines

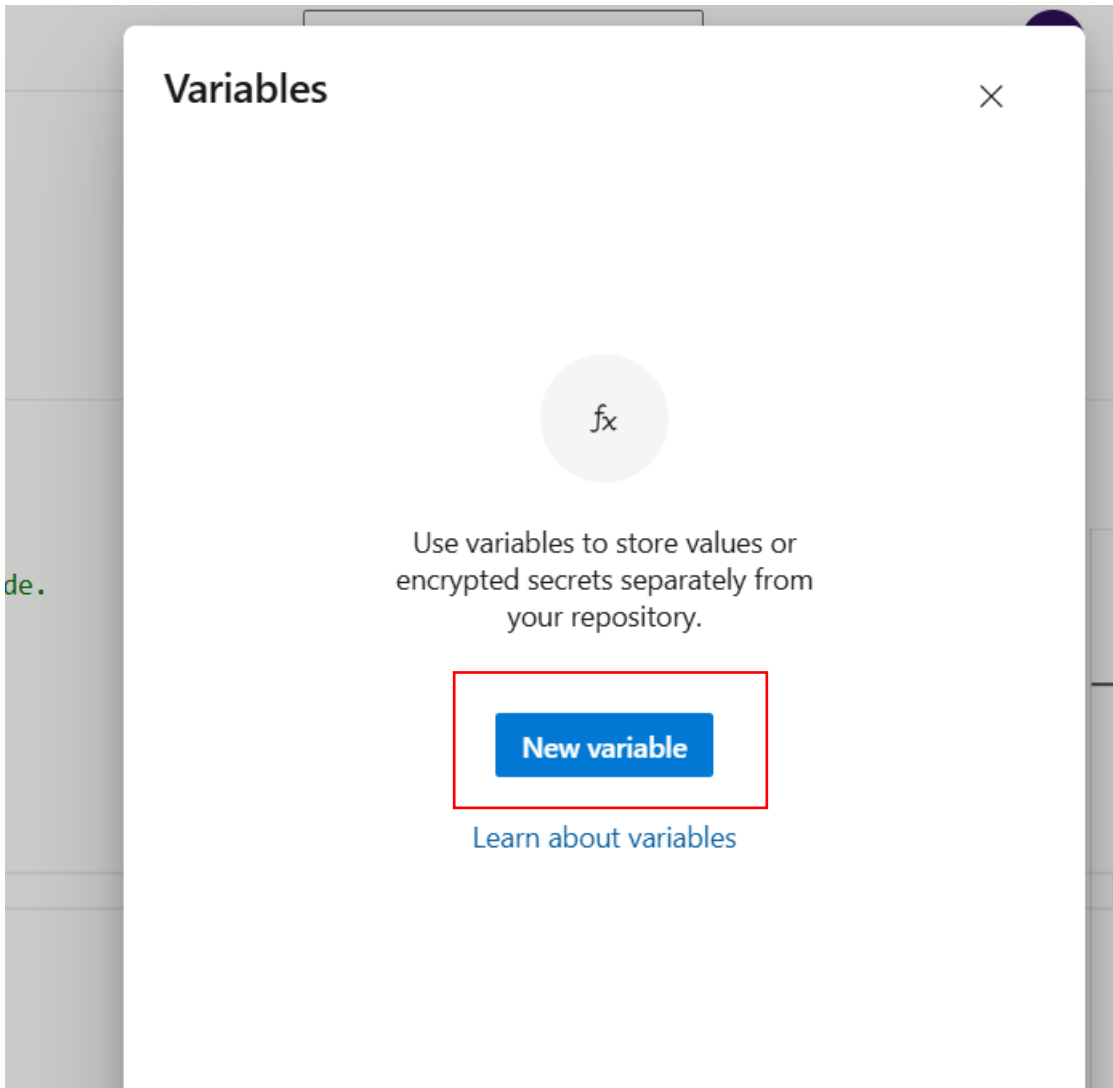
[Learn more](#)
[Troubleshoot](#)

[Back](#) [Verify and save](#) ▼

Variable Setup and Environment Pipeline

1	From the left menu option, click on Pipeline icon and select Pipelines option.  The screenshot shows the Azure DevOps interface for the 'AnitaOrganization'. The left-hand navigation menu is visible, with the 'Pipelines' option highlighted by a red rectangle. The menu items include ServiceHelpDesk, Overview, Boards, Repos, Pipelines, Environments, Library, Test Plans, and Artifacts. The 'Pipelines' option is the first item under the 'Pipelines' section.
2	Click on create pipeline button  The screenshot shows the 'Create your first Pipeline' page in Azure DevOps. The page features a large illustration of a robot and a person working on a laptop. Below the illustration, the text reads: 'Automate your build and release processes using our wizard, and go from code to cloud-hosted within minutes.' The 'Create Pipeline' button is highlighted with a red rectangle.

3	In Where is your code? Section, select the option Azure Repos Git 
4	In next page, select the repository name EnvironmentRepo 
5	Now click on Variables box

	← EnvironmentRepo Variables Run master EnvironmentRepo / azure-pipelines.yml <pre>1 # Starter pipeline 2 # Start with a minimal pipeline that you can customize to build and deploy your code. 3 # Add steps that build, run tests, deploy, and more: 4 # https://aka.ms/yaml 5 6 trigger: 7 - master 8 9 pool: 10 - name: AnitaAgentPool 11 12 steps: 13 - task: AzureResourceGroupDeployment@2 14 inputs:</pre> Tasks Search tasks .NET Core Build, test, package, or publish a dotnet applicatio... Android signing Sign and align Android APK files Ant Build with Apache Ant App Center distribute Distribute app builds to testers and users via Visu...
6	Click New Variable button 
7	In the new variable screen, enter as follows Name : app_name Value : techversant-dotnetcore-app Check the checkbox as shown below. And finally click on ok button.

Name

app_name

Value

techversant-dotnetcore-app

☐ Keep this value secret

☒ Let users override this value when running this pipeline

To reference a variable in YAML, prefix it with a dollar sign and enclose it in parentheses. For example: `$(app_name)`

To use a variable in a script, use environment variable syntax. Replace `.` and space with `_`, capitalize the letters, and then use your platform's syntax for referencing an environment variable. Examples:

Batch script: `%APP_NAME%`
PowerShell script: `${env:APP_NAME}`
Bash script: `$(APP_NAME)`

[Learn about variables](#)

Cancel

OK

Observe the variable added as shown below

Variables

×

🔍 Search variables

+

fx

app_name
= techversant-dotnetcore-app

- 8 Now click on + symbol to add variables one by one as per below table. Wherever <name> is mentioned use your name.

First row variable addition is already done in step number 7

Name	Value
app_name	techversant-dotnetcore-app
connectedservicename	Refer Appendix to get the name. Example : AnitaAzureServiceconnection
plan_name	techversant-plan
subscription_id	<Provided by ops team> Example : b4xxxxxx-exx2-4xxxd-axx-6xxxxxxxxxxxxx Not giving the full ID here due to security policy.

- 9 After entering all the variables click on save button

Variables

+

fx

app_name
= techversant-dotnetcore-app

fx

connectedservicename
= AnitaAzureServiceConnection

fx

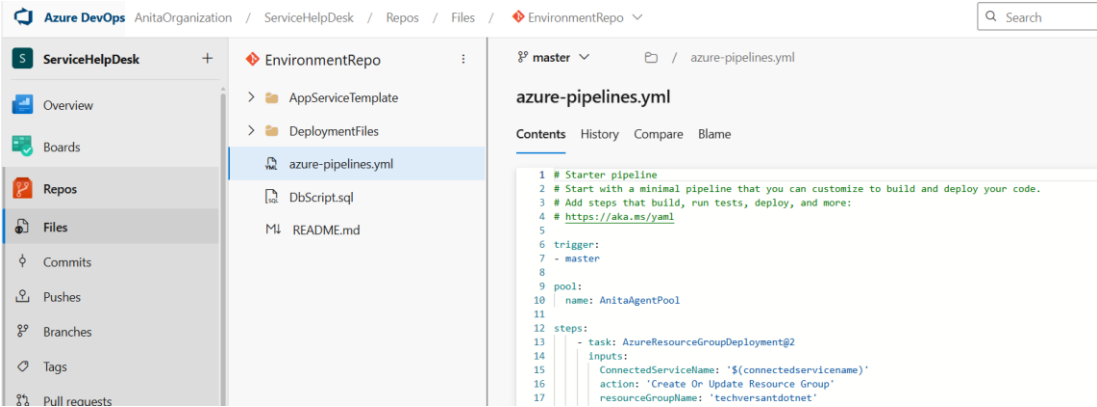
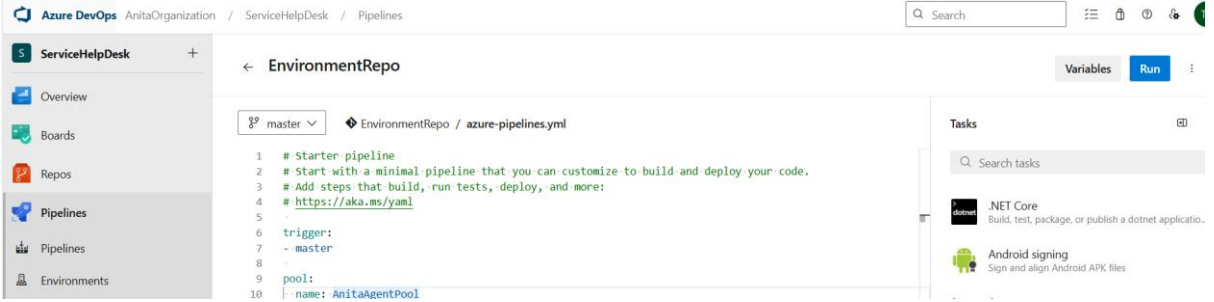
plan_name
= servicehelpdeskplananita

fx

subscription_id
= d1000be4-4abb-4675-b709-04c98058ec17

Close

[Learn about variables](#) Close Save

10	Open the file azure-pipelines.yml and click Edit button. Replace the AgentPoolName with the agent pool name which you created and click on Commit button. 
11	After saving the variable values successfully click on save and run button 
12	In Save and run window enter any message under commit message box. Here it is given updated variable values. Ensure the radio button commit to the main branch is checked and Click on Save and Run

Save and run

Saving will commit azure-pipelines.yml to the repository.

Commit message

Updated variable values

Optional extended description

Add an optional description...

☒ Commit directly to the main branch

☐ Create a new branch for this commit

Save and run

Azure DevOps

AnitaOrganization / ServiceHelpDesk / Pipelines / EnvironmentRepo / 20260109.1

ServiceHelpDesk

Overview

Boards

Repos

Pipelines

Pipelines

Environments

Library

Test Plans

Artifacts

#20260109.1 • Updated template.json

EnvironmentRepo

SummaryCode Coverage

Manually run by Test access for Anita

Repository and versionEnvironmentRepo master f4935b4b

Time started and elapsedJust now20s

Re

StagesJobs

Stage

0/1 completed15s

Job15s

Cancel

13

Click on Job link after the blue color turn to green icon

Pipelines

RecentAllRuns

Filter pipelines

Recently run pipelines

Pipeline	Last run
EnvironmentRepo	#20250215.2 • Updated parameters.json Manually triggered for main 2m ago 1m 33s

15

This execution takes 10-15 minutes of time. It varies depends on your network bandwidth. **The pipeline should be executed exactly once.** Ensure all the jobs are executed successfully.

Jobs in run #20250215.2

EnvironmentRepo

Jobs

Job1m 21s

Initialize job<1s

Checkout Environmen...19s

AzureResourceGroup...55s

Publish Artifact: Deploy...4s

Post-job: Checkout Envi...1s

Finalize Job<1s

Report build status<1s

Job

Pool: JawsanthAgentPool

Agent: agent1

Started: Today at 10:43 PM

Duration: 1m 21s

Job preparation parameters

9 queue time variables used

1 artifact produced

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If you wish to see the resources deployed, click on **show deployed resource** under the jobs which turned green and view the resource in the cmd window on the right.

Click on Pipelines from the left menu to view the EnvironmentRepo pipeline

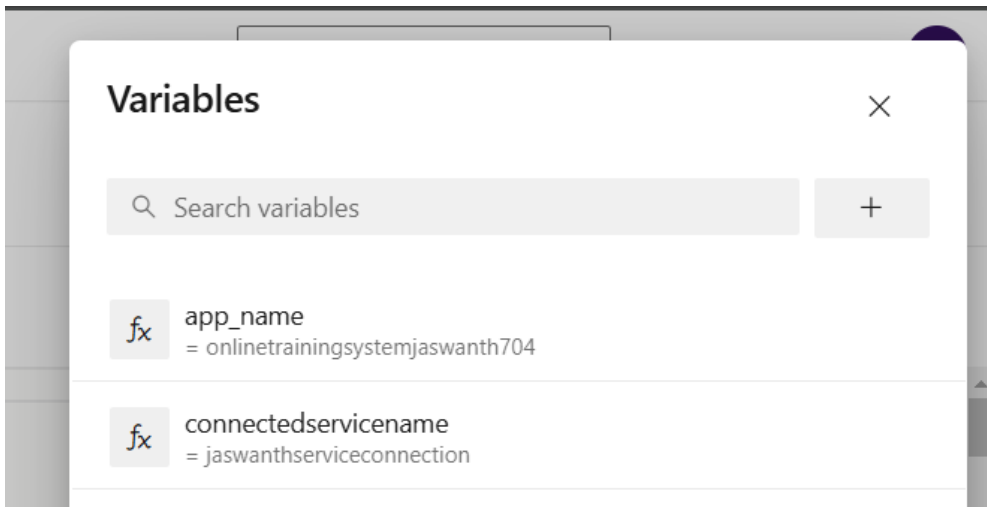
The screenshot shows the 'Pipelines' section in Azure DevOps. At the top right, there is a 'New pipeline' button and a menu icon. Below this, there are tabs for 'Recent', 'All', and 'Runs', with 'Recent' being the active tab. A 'Filter pipelines' search bar is also present. The main area displays 'Recently run pipelines' with a table. The table has two columns: 'Pipeline' and 'Last run'. One pipeline is listed: 'EnvironmentRepo' with a green checkmark icon. The 'Last run' column shows the run ID '#20210111.1', the reason 'Updated variable values', and the time '15m ago' with a clock icon. Below the run ID, it says 'Individual CI for' followed by a green circle with 'AB' and 'main'.

ServiceHelpDeskSystem Pipeline Execution

- 1 Now click on **New Pipeline** button to create one more pipeline for Deploying web application

This screenshot is similar to the previous one, showing the 'Pipelines' section. However, the 'New pipeline' button at the top right is highlighted with a red rectangular box. The 'Recently run pipelines' table shows the same 'EnvironmentRepo' pipeline, but the 'Last run' information is different: run ID '#20250215.2', reason 'Updated parameters.json', time '5m ago', and it was 'Manually triggered for' a blue circle with 'AB' on the 'main' branch.

- 2 In Where is your code? Section select **Azure Repos Git**
In Select Repository section select **ServiceHelpDeskSystem**

	New pipeline Select a repository Filter by keywords ServiceHelpDesk ▼ ✕  EnvironmentRepo  ServiceHelpDeskSystem						
3	In the displayed window, click on Variables button and add the variables similar to the previous pipeline step. <table border="1"> <thead> <tr> <th>Name</th><th>Value</th></tr> </thead> <tbody> <tr> <td>app_name</td><td>App name given in the previous section. Example : techversant-dotnetcore-app</td></tr> <tr> <td>connectedservicename</td><td>Connectedservicename given in the previous section Example : AnitaAzureserviceconnection</td></tr> </tbody> </table>	Name	Value	app_name	App name given in the previous section. Example : techversant-dotnetcore-app	connectedservicename	Connectedservicename given in the previous section Example : AnitaAzureserviceconnection
Name	Value						
app_name	App name given in the previous section. Example : techversant-dotnetcore-app						
connectedservicename	Connectedservicename given in the previous section Example : AnitaAzureserviceconnection						
4	After adding all the variables click on Save button 						
5	Open the file azure-pipelines.yml and click Edit button. Replace the AgentPoolName with the agent pool name which you created and click on Commit button						
5	Click on save and run button and enter some commit message and click save and run to trigger the pipeline						

Save and run

Saving will commit azure-pipelines.yml to the repository.

Commit message

ServiceHelpDesk Pipeline

Optional extended description

Add an optional description...

☒ Commit directly to the main branch
☐ Create a new branch for this commit

Save and run

6 Observe the pipeline stage execution started as shown below

dev.azure.com/Jaswanthorganisation/OnlineTraining/_build/results?buildId=4&view=results

Jaswanthorganisation / OnlineTraining / Pipelines / onlinetrainingsystem / 20250215.2

Search

Cancel

#20250215.2 • Updated OnlineTrainingSystem.MvcApp.csproj

onlinetrainingsystem

Summary Code Coverage

Manually run by testuser3jan

View 6 changes

Repository and version

Time started and elapsed

Related

Tests and coverage

onlinetrainingsystem

Just now

0 work items

Get started

main

cebc06f7

43s

0 artifacts

Stages Jobs

ApplicationBuild

0/1 completed 39s

Build

39s

Cancel

UnitTest

Not started

StaticCodeAnalysis

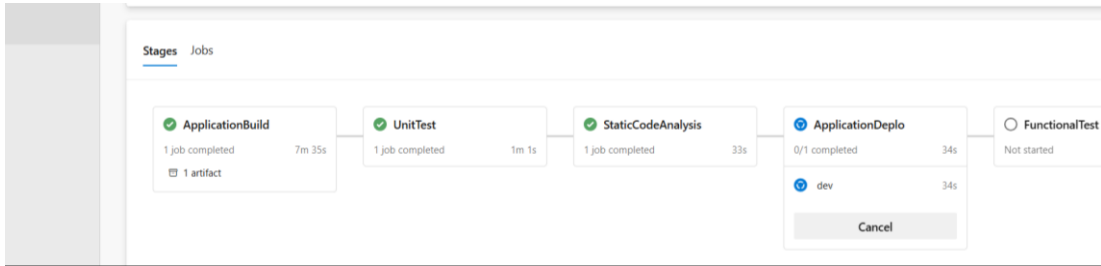
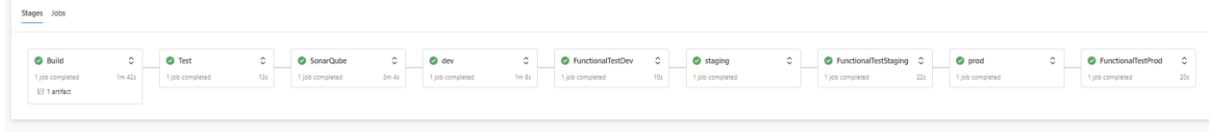
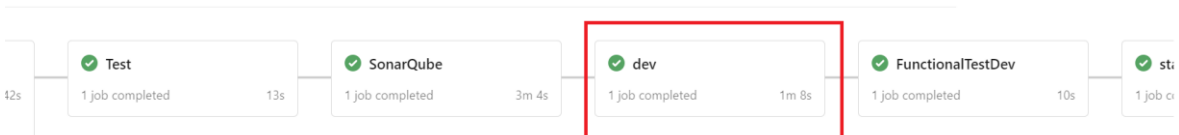
Not started

ApplicationDeplo

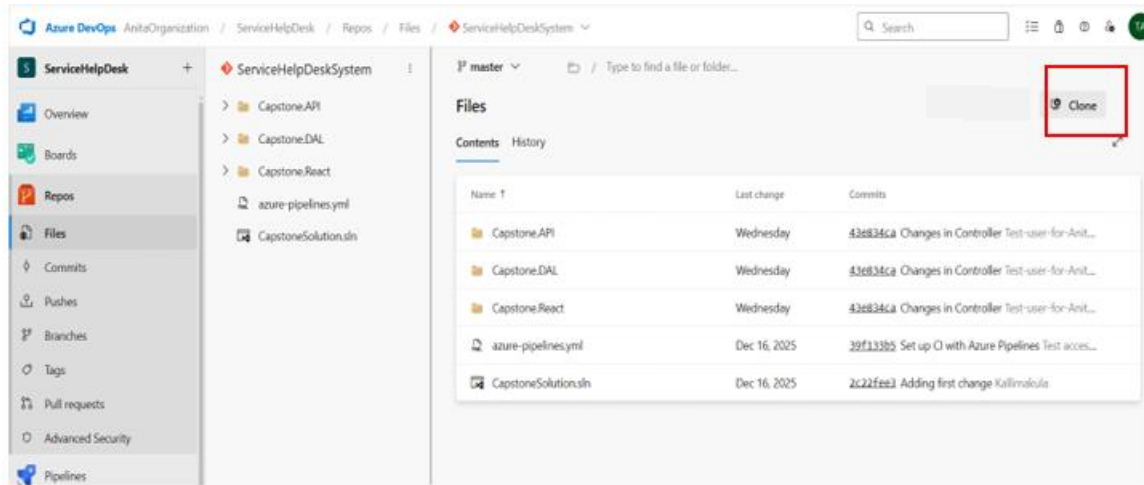
Not started

FunctionalTest

Not started

7	Observe the pipeline execution status and wait for the entire pipeline stages to complete (turn green) 
11	Complete pipeline is executed and shown below 
12	Now click on dev stage 

Clone the Repository in Visual Studio

	Go to Project Repo and Click on the Clone button 
	Copy the URL

or folder...

Last change

Wednesday

Wednesday

Wednesday

Dec 16, 202

Dec 16, 202

Clone Repository

Command line

HTTPS

SSH

https://AnitaOrganization@dev.azure.com

[Learn more about HTTPS](#)

Generate Git Credentials

IDE

Clone in VS Code

Having problems authenticating in Git? Be sure to get the latest version
① [Git for Windows](#) or our plugins for [IntelliJ](#), [Eclipse](#), [Android Studio](#) or [Windows command line](#).

Open Visual Studio click on Clone a repository Option

Visual Studio 2022

Open recent

Search recent (Alt+S)

Get started

Clone a repository

Get code from an online repository like GitHub or Azure DevOps

Open a project or solution

Open a local Visual Studio project or .sln file

Open a local folder

Navigate and edit code within any folder

Create a new project

Choose a project template with code scaffolds to get started

[Continue without code →](#)

Paste the Clone the URL and Select an Empty Folder , Click on Clone button

Clone a repository

Enter a Git repository URL

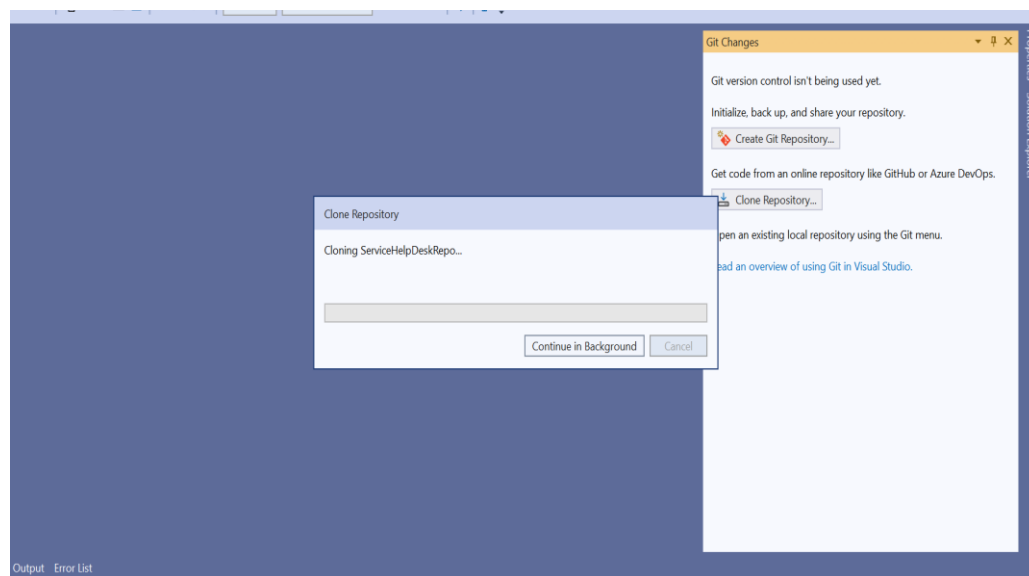
Repository location

Path

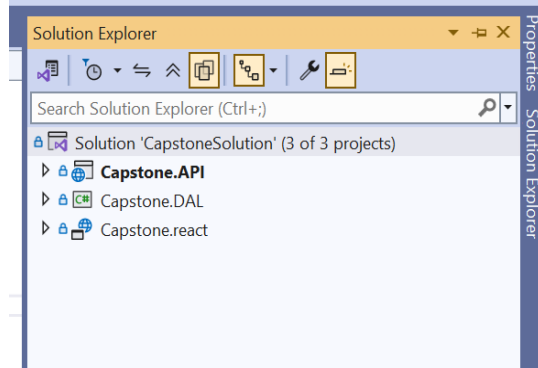
Browse a repository

-  Azure DevOps
-  GitHub

This will clone whole repository project on your local path



After Cloning Open the VS solution explorer as below

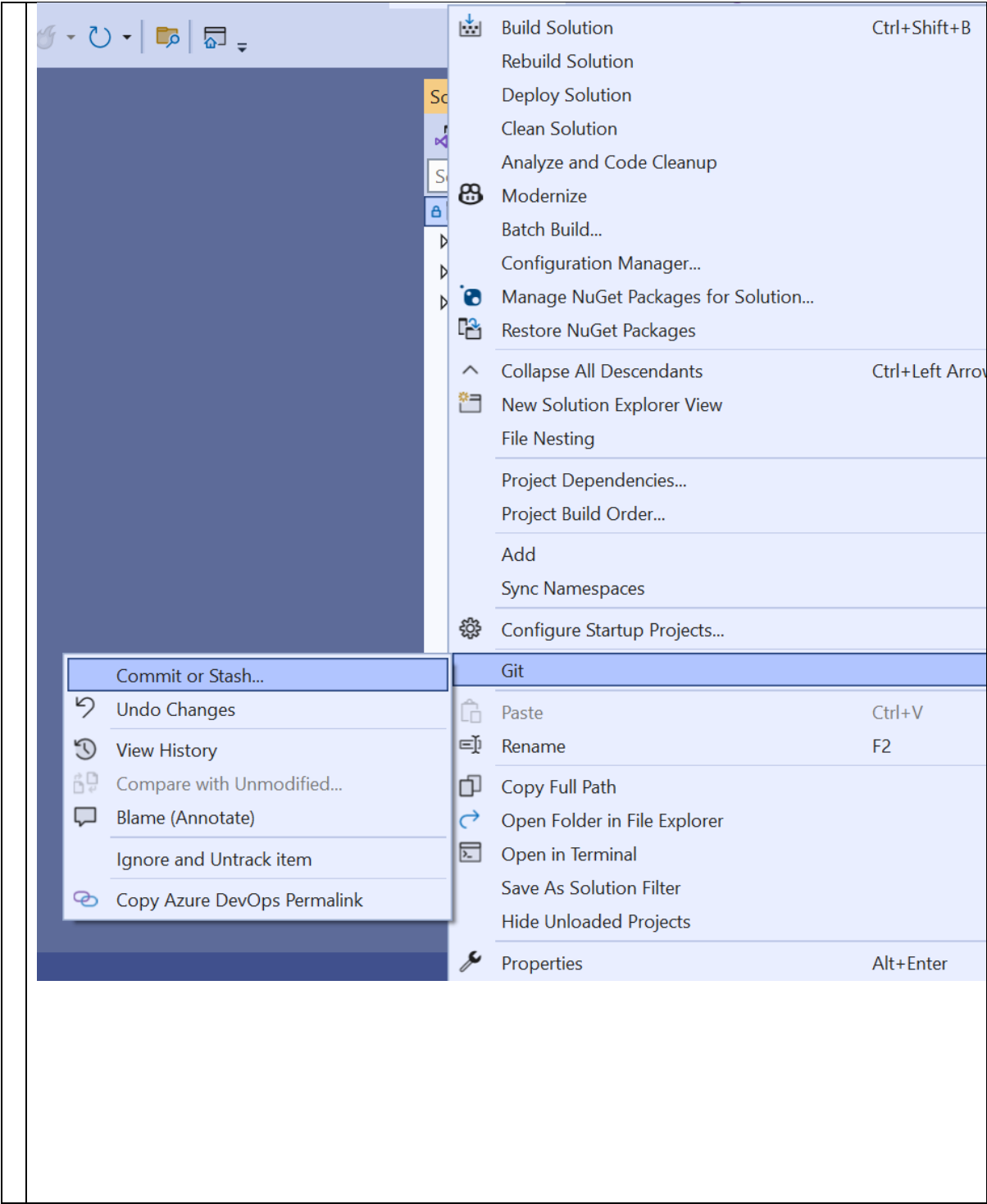


Check the visual studio must be log-in with your Azure credentials.

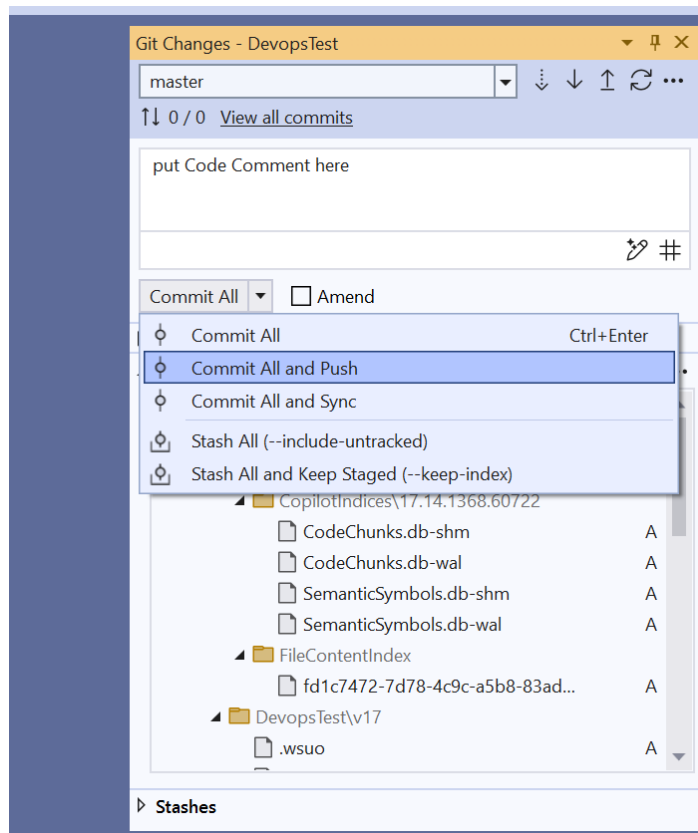
Implement the project specific code in corresponding layers as per Capstone project requirement document .

- DAL
- API
- React presentation

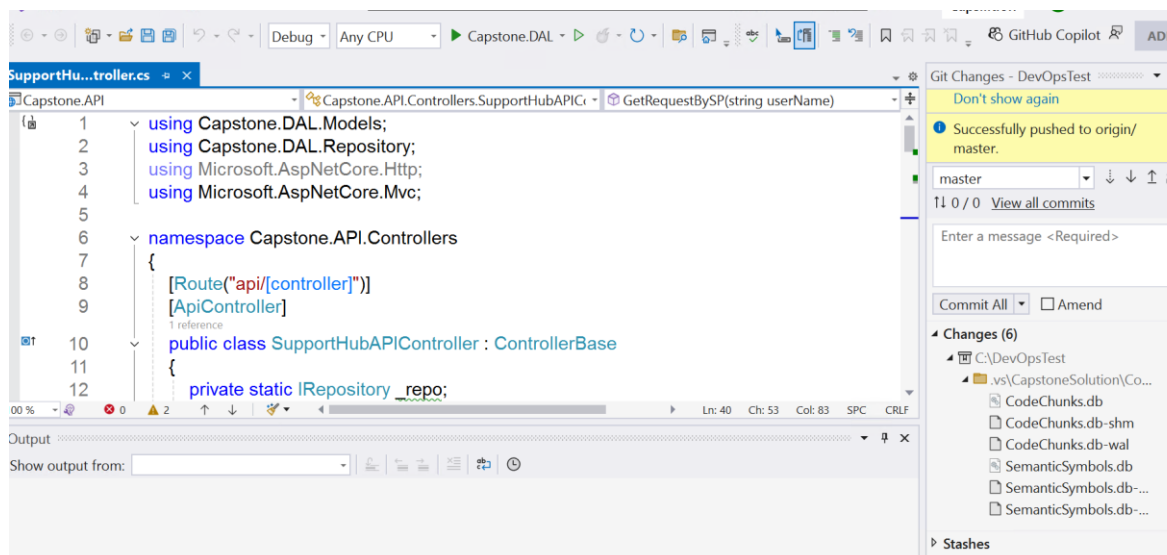
Once the implementation is done, push the changes into central repository by rightclick on Solution explorer, Select Git option



Put the proper comment as per your code and Select the option **Commit All and Push**



Once changes get pushed into Repo, it shows *successfully pushed to origin* message



Once successful commit is done from visual studio check the project repository in Azure DevOps portal. All the new changes will be seen in project Repo.

Go to Project pipeline and Run again. It should successfully run all the stages

Overview

Boards

Repos

Pipelines

Runs

Branches

Analytics

Description

Stages

#20251217.2 • all changes done

Manually run by master 6ad60241

✓-✓-✓-✓-✓(+3)

Dec 17 2025 29m

Stages

Jobs

✓ ApplicationBuild

1 job completed 8m 43s

1 artifact

✓ UnitTest

1 job completed 48s

✓ ApplicationDeplo

1 job completed 1m 10s

✓ Function

1 job completed

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APPENDIX (Optional)

To get the Client IP addresses

1. Open the new tab in the browser and go to URL <https://www.whatismyip.com/what-is-my-public-ip-address/>
2. That will show the address similar to below one. Copy the address , which is the client IP value to be entered.

The screenshot shows a web browser window at the URL <https://www.whatismyip.com/what-is-my-public-ip-address/>. The page features a navigation bar with links for 'What's My IP', 'Speed Test', 'IP Lookup', 'Change IP', and 'Hide IP'. On the left, there is a sidebar with a green 'Ask A Question' button and a list of categories: 'Questions & Answers', 'IP Tools', 'How To', 'Knowledge Base', and 'General'. The main content area displays 'Your Public IPv4 Address Is: 49.207.219.248' with the IP address highlighted in a red box. Below this, it states 'Your Public IPv6 is: Not Detected'. A banner for 'Apache Kafka Made Serverless' by Confluent Cloud is visible at the top right. The section is titled 'What Is My Public IP Address?' and includes a paragraph explaining that the public IP address is the Internet Protocol address logged by various servers/devices, which is the same IP address shown on the homepage.